

Pattern Recognition and Machine Learning (EE602L)

Lab 2: Regression — Parameter Estimation and Regularization

Instructions:

Use Google Colab.

Submit one well-commented notebook (Lab2-RollNum.ipynb) along with a PDF report (Lab2-RollNum.pdf) containing the experimental outputs, analysis of the results, and conclusions.

Wrong file names or missing files won't be graded.

Deadline: Monday, 24 August, 11:59 PM. Late submissions lose 10% per day.

This is an individual assignment. Please follow the Institute's Academic Code of Honor.

Dataset:

housing.csv (California Housing dataset, 1990 U.S. Census)

Source: <https://raw.githubusercontent.com/ageron/handson-ml2/master/datasets/housing/housing.csv>

Data Preparation (common to both parts):

Before starting either part, load housing.csv and get it ready for modelling. Inspect the shape, column types and summary statistics, handle missing values, duplicates and any obvious outliers (state and consistently apply one clear criterion for outliers, e.g., removal, capping, or flagging), and encode the categorical column (ocean_proximity). Visualize the data before you model it: at minimum, look at the distribution of the target and how it relates to a couple of features. Then implement a 70:15:15 train-validation-test split on your own (NumPy/Pandas only, no sklearn.model_selection), check that the split sizes are correct and that no row appears in more than one set, and standardize the numerical features, impute missing values, and decide on outliers using statistics/criteria computed from the training set only.

Carry this same cleaned, split and scaled dataset through both Part (a) and Part (b) below. You should not need to repeat any of this work.

Part (a): MMSE Parameter Estimation vs. Gradient Descent

Add an intercept term and estimate the regression parameters using the closed-form MMSE solution $\theta = (X^T X)^{-1} X^T y$, using a numerically safer pseudo-inverse where needed and explaining why; write a predict function $\hat{y} = X\theta$, compute training and validation error using $MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$, $RMSE = \sqrt{MSE}$, $MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$, $R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$ and $R_{adj}^2 = 1 - (1 - R^2) \frac{n-1}{n-p-1}$, (p is the number of predictors used, excluding the intercept), and report the learned intercept and coefficients with their feature names. Then implement gradient descent from scratch using the update rule $\theta \leftarrow \theta - \alpha \nabla_{\theta} J(\theta)$ with $\nabla_{\theta} J(\theta) = -\frac{2}{n} X^T (y - X\theta)$ for the MSE cost $J(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$; train it with at least three learning rates, plot the training-cost curves for comparison, and pick the best setting using validation MSE. Modify this implementation to optimize the MAE cost $J(\theta) = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$ instead, using the valid subgradient $\frac{\partial}{\partial \hat{y}_i} |y_i - \hat{y}_i| = \text{sign}(\hat{y}_i - y_i)$, taken as any value in $[-1, 1]$ when $y_i = \hat{y}_i$ (equivalently, the full gradient is $\nabla_{\theta} J_{MAE}(\theta) = -1/n \cdot X^T \text{sign}(y - X\theta)$, applied element-wise with $\text{sign}(0)$ taken as any value in $[-1, 1]$), and compare the MSE-loss and MAE-loss models in a compact validation-metrics table, briefly explaining the trade-off. Finally, evaluate both the MMSE solution and the best MSE-cost gradient-descent model on the test set, present a compact comparison table (training time for that single best-learning-rate run, test MSE, RMSE, MAE, R^2 , R_{adj}^2 , and the L2 norm of the difference between their coefficients, $\|\theta_1 - \theta_2\|_2$, excluding the intercept), plot actual-vs-predicted values for the better model, and conclude in a few sentences which approach is preferable for a dataset of this size, which would scale more naturally to much larger data, and why.

Part (b): Linear Regression vs. Ridge vs. Lasso

Using the same prepared features, fit a plain (unregularized) Linear Regression model (you may reuse your Part (a) MMSE solution), then examine the numerical features' correlation matrix and the condition number of the training feature matrix to identify at least two pairs or groups of features that are highly correlated with each other (for instance, total_rooms, total_bedrooms, population and households, which all tend to scale with the size of a district), and comment on what this multicollinearity does to the stability of the fitted coefficients. Implement Ridge Regression using the closed-form solution $\theta_{ridge} = (X^T X + \alpha R)^{-1} X^T y$ where $R = \text{diag}(0, 1, \dots, 1)$ is the diagonal regularization matrix, so that the intercept term is not regularized, sweep at least five values of $\alpha > 0$ along with $\alpha = 0$ as the unregularized baseline, and select the best

positive α using validation MSE. Separately fit Lasso Regression, which solves $\hat{\theta}_{lasso} = \arg \min_{\theta} \left(\frac{1}{2n} \|y - X\theta\|_2^2 + \alpha \|\theta\|_1 \right)$ using `sklearn.linear_model.Lasso` (with `fit_intercept=True`; `sklearn` does not regularize the intercept) over the same positive range of α values (excluding $\alpha = 0$), again selecting the best via validation MSE. Note that because the Ridge objective above is unnormalized while the Lasso objective is normalized by $2n$, the same numeric alpha does not represent the same regularization strength in both models; sharing an alpha range is intentional here, and you should discuss this scale difference when comparing the two coefficient paths rather than expecting them to align at equal alpha. Report how many coefficients Lasso drives exactly to zero and which features they correspond to, along with a short explanation of why Lasso can produce exact zeros while Ridge cannot. Plot coefficient magnitude against α for both Ridge and Lasso side by side and briefly describe how the two paths differ, then evaluate the plain Linear Regression baseline, the best Ridge model and the best Lasso model on the untouched test set in one compact comparison table (test MSE, RMSE, MAE, R^2 , R^2_{adj} , coefficient L2 norm $\|\theta\|_2$ (excluding the intercept), and number of non-zero coefficients). Close with a short-written discussion, grounded in your specific results, of when you would prefer plain Linear Regression, when Ridge, and when Lasso, including one real-world scenario outside this dataset where each of Ridge's and Lasso's behaviour would be especially useful.